ALLEN CAREER INSTITUTE
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KOTA (RAJASTHAN)



INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Metallurgy

English Medium

GENERAL PRINCIPLES & PROCESSES OF ISOLATION OF ELEMENTS

EXERCISE # O-1

ONLY ONE OPTION IS CORRECT.

ORES

1. Which of the following does not contain Mg:

(A) magnetite (B) magnesite (C) asbestos (D) carnallite

ML0001

2. Which of the following is not an ore:

(A) malachite (B) calamine (C) stellite (D) cerussite

ML0002

3. Carnallite does not contain

(A) K (B) Ca (C) Mg (D) Cl

ML0003

4. Among the following statements, the incorrect one is

(A) calamine and siderite are carbonate ores (B) argentite and cuprite are oxide ores

(C) zinc blende and pyrites are sulphide ores (D) malachite and azurite are ores of copper

ML0004

5. Select the correct statement :

(A) Magnetite is an ore of manganese (B) Pyrolusite is an ore of lead

(C) Siderite is carbonate ore of iron (D) FeS_2 is rolled gold

ML0005

6. "Fool's gold" is

(A) iron pyrites (B) horn silver (C) copper pyrites (D) bronze

ML0006

7. **Assertion :** Platinum and gold occur in native state in nature.

Reason : Platinum and gold are noble metals.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

ML0007

CONCENTRATION METHODS

8. $Ag_2S + NaCN + Zn \longrightarrow Ag$

This method of extraction of Ag by complex formation and then its displacement is called:

(A) Parke's method (B) McArthur-Forest method

(C) Serpeck method (D) Hall's method

ML0008

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9. Which one of the following is not a method of concentration of ore?

(A) gravity separation (B) froth floating process

(C) electromagnetic separation (D) smelting

ML0009

10. Chemical leaching is useful in the concentration of:

(A) copper pyrites (B) bauxite (C) galena (D) cassiterite

ML0010

11. In froth-floatation process, pine oil functions as

(A) activator (B) frother (C) collector (D) agitator

ML0011

12. Collectors are the substances which help in attachment of an ore particle to air bubble in froth. A popular collector used industrially is

(A) sodium ethyl xanthate (B) sodium xenate

(C) sodium pyrophosphate (D) sodium nitroprusside

ML0012

13. In the cyanide process involving extraction of silver, zinc is used industrially as a(an)

(A) oxidising agent (B) reducing agent

(C) solvent (D) solvating agent

ML0013

14. During initial treatment, preferential wetting of ore by oil and gangue by water takes place in

(A) Levigation (gravity separation) (B) Froth floatation

(C) Leaching (D) Bessemerisation

ML0014

15. An non-magnetic ore containing the impurity of $FeCr_2O_4$ is concentrated by

(A) magnetic-separation (B) gravity separation

(C) froth-floatation method (D) electrostatic method

ML0015

16. The beneficiation of the sulphide ores is usually done by

(A) Electrolysis (B) Smelting process

(C) Metal displacement method (D) Froth flotation method

ML0016

17. The process of the isolation of a metal by dissolving the ore in aqueous solution of suitable chemical reagent followed by precipitation of the metal by a more electropositive metal is called:

(A) hydrometallurgy (B) electrometallurgy

(C) zone refining (D) electrorefining

ML0017

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18. Froth floatation process for concentration of ores is an illustration of the practical application of:

(A) Adsorption (B) Absorption

(C) Coagulation (D) Sedimentation

ML0018

19. **Assertion :** Sulphide ores are concentrated by froth floatation process.

Reason : Pine oil acts as a frothing agent in froth floatation process.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

ML0019

20. **Assertion :** Wolframite impurities are separated from cassiterite by electromagnetic separation.

Reason : Cassiterite being magnetic is attracted by the magnet and forms a separate heap.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

ML0020

CALCINATION/ROASTING

21. Calcination is the process of heating the ore:

(A) in inert gas (B) in the presence of air

(C) in the absence of air (D) in the presence of CaO and MgO

ML0021

22. When roasting is carried out :

(i) Sulphide ore is converted into oxide and sulphate

(ii) remove water of hydration

(iii) the ore melts

(iv) arsenic and sulphur impurities are removed

Of these statements:

(A) (i), (ii) and (iii) are correct (B) (i) and (iv) are correct

(C) (i), (ii) and (iv) are correct (D) (ii), (iii) and (iv) are correct

ML0022

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REDUCTION PROCESS

23. In the alumino thermite process, Al acts as

(A) An oxidising agent (B) A flux

(C) A reducing agent (D) A solder

ML0023

24. **Assertion :** Al is used as a reducing agent in aluminothermy.

Reason : Al has a lower melting point than Fe, Cr and Mn.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

ML0024

25. Formation of metallic copper from the sulphide ore in the commercial thermo-metallurgical process essentially involves which one of the following reaction:

(A) $Cu_2S + \frac{3}{2}O_2 \longrightarrow Cu_2O + SO_2$; $CuO + C \longrightarrow Cu + CO$

(B) $Cu_2S + \frac{3}{2}O_2 \longrightarrow Cu_2O + SO_2$; $2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$

(C) $Cu_2S + 2O_2 \longrightarrow CuSO_4$; $CuSO_4 + Cu_2S \longrightarrow 3Cu + 2SO_2$

(D) $Cu_2S + \frac{3}{2}O_2 \longrightarrow Cu_2O + SO_2$; $Cu_2O + CO \longrightarrow 2Cu + CO_2$

ML0025

26. The element which could be extracted by electrolytic reduction of its oxide dissolved in a high temperature melt is:

(A) sodium (B) magnesium (C) fluorine (D) aluminium

ML0026

27. In which of the following isolations no reducing agent is required:

(A) iron from haematite (B) Tin from cassiterite

(C) mercury from cinnabar (D) zinc from zinc blende

ML0027

PURIFICATION METHODS

28. A metal has a high concentration into the earth crust and whose oxides cannot be reduced by carbon. The most suitable method for the extraction of such metal is

(A) Alumino thermite process (B) Electrolysis process

(C) Van-Arkel's process (D) Cupellation

ML0028

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29. **Assertion :** Alkali metals can not be prepared by the electrolysis of their chlorides in aqueous solution
Reason : Reduction potentials of alkali metals cations is much lower than that of H_2O.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

ML0029

30. **Assertion :** Magnesium can be prepared by the electrolysis of aq. $MgCl_2$.
Reason : The reduction potential of Mg^{2+} is much lower than that of H_2O.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

ML0030

31. Bessemerisation is carried out for

I : Fe,	II : Cu,	III : Al,	IV : silver
(A) I , II	(B) II, III	(C) III, IV	(D) I, III

ML0031

32. In the extraction of nickel by Mond process, the metal is obtained by:
(A) electrochemical reduction (B) thermal decomposition
(C) chemical reduction by aluminium (D) reduction by carbon

ML0032

33. Formation of $Ni(CO)_4$ and subsequent its decomposition into Ni and CO (recycled) makes basis of Mond's process

$$Ni + 4CO \xrightarrow{T_1} Ni(CO)_4 \xrightarrow{T_2} Ni + 4CO$$

T_1 and T_2 are:
(A) 100°C, 50°C (B) 50°C, 100°C (C) 50°C, 230°C (D) 230°C, 50°C

ML0033

34. Zone refining is based on the principle of
(A) fractional distillation (B) fractional crystallisation
(C) partition coefficient (D) chromatographic separation

ML0034

35. Si and Ge used for semiconductors are required to be of high purity and hence purified by
(A) zone-refining (B) electrorefining
(C) Van-Arkel's process (D) cupellation process

ML0035

36. Which process of purification is represented by the following equation :

$$Ti\ (Impure) + 2I_2 \xrightarrow{250°C} TiI_4 \xrightarrow{1400°C} Ti\ (Pure) + 2I_2$$

(A) Cupellation (B) Poling (C) Van-Arkel Process (D) Zone refining

ML0036

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37. Which of the following employ(s) thermal decomposition of volatile iodide compounds?
(A) Thermite process (B) Hall's process (C) Van-Arkel's process (D) Mond's process

ML0037

38. The method of zone refining of metals is based on the principle of:
(A) Greater mobility of the pure metal than that of impurity.
(B) Higher melting point of the impurity than that of the pure metal.
(C) Greater noble character of the solid metal than that of the impurity
(D) Greater solubility of the impurity in the molten state than in the solid

ML0038

39. **Assertion :** Titanium is purified by Van-Arkel method.
Reason : Ti reacts with I_2 to form volatile TiI_4 which decomposes at 1673 K to give pure Ti.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

ML0039

40. **Assertion :** Nickel is purified `by the thermal decomposition of nickel tetracarbonyl.
Reason : Nickel is a transition element.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

ML0040

41. Refining of silver is done by:
(A) liquation (B) poling (C) cupellation (D) van Arkel method

ML0041

42. Mercury is purified by:
(A) Passing through dilute HNO_3 (B) Distillation
(C) Distribution (D) Vapour phase refining

ML0042

43. **Assertion :** Lead, tin and bismuth are purified by liquation method.
Reason : Lead, tin and bismuth have low m.p. as compared to impurities.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

ML0043

44. When an impurity in a metal has greater affinity for oxygen and is more easily oxidised than the metal itself. Then, the metal is refined by
(A) cupellation (B) zone-refining (C) distillation (D) electrolytic process

ML0044

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EXTRACTION OF METALS

45. Which of the following process is not associated with recovery of the silver -

(A) As a side product in electrolytic refining of copper

(B) Parke's process in which Zn is used to extract silver by solvent extraction from molten lead

(C) By reaction of silver sulphide with KCN and then reaction of soluble complex with Zn

(D) By boiling $Na[Ag(CN)_2]$ aq.

ML0045

46. Blister Cu is about:

(A) 60% Cu (B) 90% Cu (C) 98% Cu (D) 100% Cu

ML0046

47. Iron obtained from blast furance is:

(A) wrought iron (B) cast iron (C) pig iron (D) steel

ML0047

48. Which of the following term is not related to Al-extraction

(A) Serpek's process (B) Hall-Heroult process

(C) Thermite process (D) Hoop's process

ML0048

49. Dow's process

(A) involves purification of copper (B) involves extraction of magnesium

(C) gives metal chloride as product (D) gives pure Na as product

ML0049

50. Silica is added to roasted copper ores during extraction in order to remove

(A) cuprous sulphide (B) ferrous oxide (C) ferrous sulphide (D) cuprous oxide

ML0050

51. Addition of high proportions of manganese makes steel useful in making rails of railroads, because manganese

(A) gives hardness to steel (B) helps the formation of oxides of iron

(C) can remove oxygen and sulphur (D) can show highest oxidation state of +7

ML0051

52. In the commercial electrochemical process for aluminium extraction the electrolyte used is

(A) $Al(OH)_3$ in NaOH solution (B) an aqueous solution of $Al_2(SO_4)_3$

(C) a molten mixture of Al_2O_3 ,Na_3AlF_6 & CaF_2 (D) a molten mixture of Al_2O_3 and $Al(OH)_3$

ML0052

53. Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases (like CH_4). This process X is called _____ and the metal contains impurities of Y is _____.

(A) X = cupellation, Y = CuO_2 (B) X = poling, Y = Cu_2O

(C) X = poling, Y = CuO (D) X = cupellation, Y = CuO

ML0053

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54. A piece of steel is heated until redness and then plunged into cold water or oil. This treatment of steel makes it
(A) soft and malleable
(B) hard but not brittle
(C) more brittle
(D) hard and brittle

ML0054

55. Modern method of steel manufacturing is
(A) open hearth process
(B) L.D. Process
(C) Bessemerisation
(D) Cupellation

ML0055

56. During electrolytic reduction of alumina, two auxiliary electrolytes X and Y are added to increase the electrical conductance and lower the temperature of melt in order to making fused mixture very conducting. X and Y are
(A) cryolite and flourspar
(B) cryolite and alum
(C) alum and flourspar
(D) flourspar and bauxite

ML0056

57. For extraction of sodium from NaCl, the electrolytic mixture NaCl + KCl + $CaCl_2$ is used. During extraction process, only sodium is deposited on cathode but K and Ca do not because
(A) Na is more reactive than K and Ca
(B) Na is less reactive than K and Ca
(C) NaCl is less stable than Na_3AlF_6 and $CaCl_2$
(D) the discharge potential of Na^+ is less than that of K^+ and Ca^{2+} ions.

ML0057

58. Railway wagon axles are made by heating iron rods embedded in charcoal powder. This process is known as:
(A) Sherardising (B) Annealing (C) Tempering (D) Case hardening

ML0058

59. In the extraction of copper from its sulphide ore the metal is formed by the reduction of Cu_2O with:
(A) FeS (B) CO (C) Cu_2S (D) SO_2

ML0059

60. Carnallite on electrolysis gives:
(A) Ca and Cl_2 (B) Na and CO_2 (C) Al and Cl_2 (D) Mg and Cl_2

ML0060

MISCELLANEOUS

61. Which of the following statement is correct regarding Cu-extraction
(A) In the smelting step carbon reduction takes places
(B) During partial roasting Cu_2S remains almost unaffected
(C) In Bessemer converter, only self reduction occur, not slag formation
(D) Blister forms in the blister Cu is due to dissolved CO_2

ML0061

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62. Refractory materials are generally used in furnaces because

(A) they are chemically inert (B) they can withstand high temperature

(C) they do not contain impurities (D) they decrease melting point of ore

ML0062

63. Which of the following statements is correct regarding the slag formation during the extraction of a metal like copper or iron.

(A) The slag is lighter and has lower melting temperature than the metal

(B) The slag is heavier and has lower melting temperature than the metal

(C) The slag is lighter and has higher melting temperature than the metal

(D) The slag is heavier and has higher melting temperature than the metal

ML0063

64. **Assertion :** Generally in smelting, roasted/cacinated ore is heated with powdered coke in presence of a flux.

Reason : Oxides are reduced to metals by C or CO. Impurities are removed as slag.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

ML0064

65. **Assertion :** Magnesia and quick lime are used as basic flux.

Reason : MgO and CaO can withstand very high temperatures.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

ML0065

66. **Assertion :** Wolframite impurity is separated from SnO_2 by magnetic separation

Reason : Tin stone is ferromagnetic, therefore attracted by magnet.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

ML0066

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EXERCISE # O-2

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

1. Which of the following is(are) sulphide ores?
(A) Argentite (B) Galena (C) Anglesite (D) Copper glance

ML0067

2. Which of the following is (are) regarded as iron ores?
(A) Haematite (B) Magnetite (C) Limonite (D) Copper pyrites

ML0068

CONCENTRATION

3. Which of the following ores is(are) concentrated by froth floatation?
(A) haematite (B) galena (C) copper pyrite (D) azurite

ML0069

4. Which of the following ores is (are) concentrated industrially by froth floatation?
(A) Copper pyrites (B) Galena (C) Dolomite (D) Carnallite

ML0070

5. Leaching is used for the concentration of:
(A) Red bauxite (B) Haematite (C) Gold ore (D) Silver ore

ML0071

CALCINATION/ROASTING

6. Calcination and roasting processes of ores to form their oxides are beneficial
(A) to convert ores into porous form so that their reduction becomes easier
(B) as impurities like S, As, Sb, are removed
(C) as organic impurities are removed.
(D) as the ores are converted into oxide form which makes the reduction easier

ML0072

7. Which of the following reaction(s) occur during calcination?
(A) $CaCO_3 \rightarrow CaO + CO_2$ (B) $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$
(C) $2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$ (D) $CuS + CuSO_4 \rightarrow 2Cu + 2SO_2$

ML0073

8. Which of the following is true for calcination of a metal ore?
(A) It makes the ore more porous
(B) The ore is heated to a temperature when fusion just begins
(C) Hydrated salts lose their water of crystallisation
(D) Impurities of S, As and Sb are removed in the form of their volatile oxides.

ML0074

9. Roasting can be performed in
(A) blast furnace (B) reverberatory furnace
(C) electric furnace (D) None of these

ML0075

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REDUCTION

10. Carbon reduction method is employed for commercial extraction of metal from amongst these :
(A) haematite (B) cassiterite (C) iron pyrite (D) corundum

ML0076

11. Auto reduction process is used in extraction of
(A) Cu (B) Hg (C) Al (D) Fe

ML0077

12. Which of the following reduction reactions are actually employed in commerical extraction of metals?
(A) $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$
(B) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$
(C) $2Na[Au(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Au$
(D) $Cu_2S + Pb \rightarrow Cu + PbS \downarrow$

MΛ0078

PURIFICATION

13. In the manufacturing of metallic sodium by fused salt-electrolysis method (Down's process), small amount of $CaCl_2$ that added is known as auxiliary electrolyte and is used to
(A) improve the electrical conductance (B) decrease the melting point of electrolyte
(C) stabilise the metallic sodium (D) increase the temperature of electrolysis

ML0079

14. Poling is employed in refining of
(A) iron (B) copper (C) tin (D) lead

ML0080

15. Zone refining is used for purification of
(A) Ge (B) Si (C) Ga (D) In

ML0081

16. Metal(s) which does/do not form amalgam is/are
(A) Fe (B) Pt (C) Zn (D) Au

ML0082

17. Metals which can be commercially extracted by smelting process
(A) Pb (B) Fe (C) Zn (D) Mg

ML0083

EXTRACTION OF METALS

18. Hoop's process of purification of aluminium involves formation of layers during electrolysis. It involves
(A) the three layers have same densities but different materials.
(B) the three layers have different densities
(C) the upper layer is of pure aluminium which acts as a cathode
(D) the bottom layer is of impure aluminium which acts as an anode and middle layer consists of cryolite and BaF_2.

ML0084

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19. Metallurgical process of zinc involves roasting of zinc sulphide followed by reduction. Metallic zinc distills over as it is volatile and impurities like Cu, Pb and Fe gets condensed. The crude metal obtained is called spelter, which may be purified by

(A) electrolysis process (B) fractional distillation

(C) polling (D) heating with iodine

ML0085

20. Which of the following process (es) are used for purification of Bauxite ore?

(A) Hall's process (B) Serpeck's process (C) Baeyer's process (D) Mond's process

ML0086

21. Common impurities present in Bauxite are

(A) CuO (B) ZnO (C) Fe_2O_3 (D) SiO_2

ML0087

22. Calcium silicate slag formed in extraction of iron

(A) prevents the reoxidation of molten iron.

(B) catalyses the combustion of carbon.

(C) reduces CO_2 to CO at the bottom of the furnace.

(D) is used in cement industry.

ML0088

23. Amphoteric nature of aluminium is employed in which of the following process for extraction of aluminium?

(A) Baeyer's process (B) Hall's process

(C) Serpec's process (D) Dow's process

ML0089

24. The chief reaction(s) occuring in blast furnace during extraction of iron from haematite is(are)

(A) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ (B) $FeO + SiO_2 \rightarrow FeSiO_3$

(C) $Fe_2O_3 + C \rightarrow 2Fe + 3CO$ (D) $CaO + SiO_2 \rightarrow CaSiO_3$

ML0090

25. Which of the following are true for electrolytic extraction of aluminium

(A) cathode material contains graphite

(B) anode material contains graphite

(C) cathode reacts away forming CO_2

(D) anode reacts away forming CO_2

ML0091

26. During extraction of copper, it is obtained in the form of molten *matte*. Which of the following is **not true**?

(A) *matte* is further treated in Bessemer's coverter

(B) molten *matte* is electrolysed

(C) It is treated with a blast of air and sand

(D) It is dissolved in $CuSiF_6$ and crystallised.

ML0092

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27. The major role of fluorspar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is

(A) as a catalyst

(B) to make the fused mixture very conducting

(C) to lower the melting temperature of the mixture

(D) to decrease the rate of oxidation of carbon at the anode.

ML0093

28. Which of the following reaction does not occur in blast furance during extraction of iron :

(A) $CaO + SiO_2 \longrightarrow CaSiO_3$

(B) $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$

(C) $FeO + SiO_2 \longrightarrow FeSiO_3$

(D) $FeO \longrightarrow Fe + \frac{1}{2}O_2$

ML0094

MISCELLANEOUS

29. Which of the following employ downward movement of ore due to gravity?

(A) Gravity separation

(B) Froth floatation

(C) Blast furnace

(D) Bessemer's coverter

ML0095

30. The **CORRECT** statements are :

(A) generally the calcination and roasting is done in blast furance

(B) the sandy and rocky materials associated with ore are called matrix

(C) froth floatation process is suitable for sulphide ores

(D) substance that reacts with gangue to form fusible mass is called slag

ML0096

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EXERCISE # S-1

1. Find the number of ore which are concentrated by magnetic sepration method.
Haemetite, Cassiterite, Copper Glance, Chromite, Cinnabar

ML0097

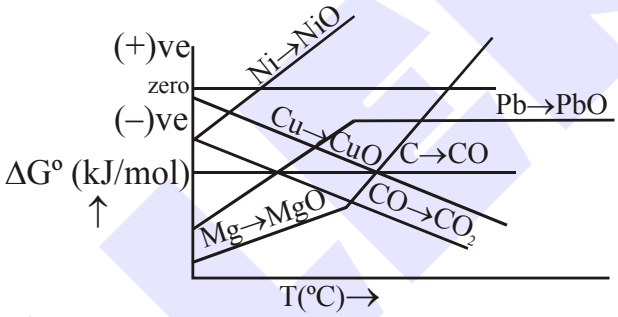
2. Find the number of metals which are commercially extracted by carbon reduction method
Pb, Fe, Zn, Mg, Al, Na, Au, Ag

ML0098

3. The number of following pairs is correctly matched
(i) Van Arkel method – Zirconium
(ii) Mond Process – Titanium
(iii) Froth Floatation Method – Cerussite
(iv) Distillation method – Zinc
(v) Poling Process – Copper
(vi) amalgamation – Gold

ML0099

4. Find the number of curves which are wrongly presented in the Ellingham diagram.



ML0100

5. How many of the following minerals containing Mg.
Magnetite, Carnallite, Epsom salt, Siderite

ML0101

6. Find out the number of minerals given below contain iron as Fe(II).
Haematite, Magnetite, Limonite, Siderite, Chromite, Wolframite

ML0102

7. Amongst the following ores, the total number of oxide ores are
Siderite, Magnetite, Haematite, Malachite, Zincite, Cuprite

ML0103

8. Amongst the following, total number of sulphide ores are
Calamine, Sphalertie, Copper pyrites, Copper glance, Iron pyrites, Bauxite

ML0104

9. How many of the following ores of silver ?
Hornsilver, Cerrusite, Chalcopyrite, Galena, Anglesite, Argentite

ML0105

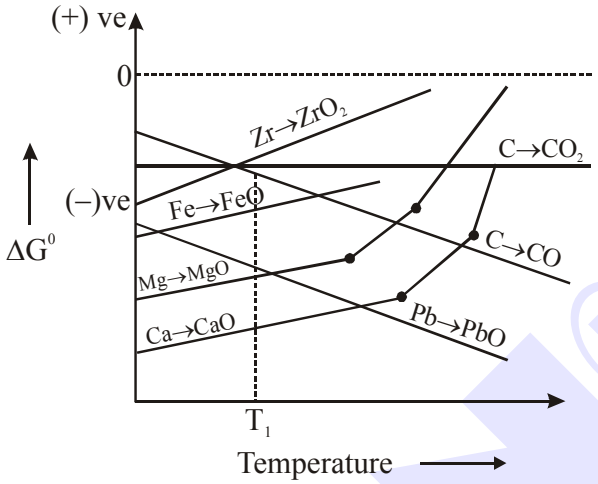
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EXERCISE # S-2

COMPREHENSION AND MATCH THE COLUMN

ELLINGHAM DIAGRAM

Paragraph for 1 to 3



1. Which of the above curve is wrongly presented -

(A) $C \rightarrow CO_2$ (B) $Pb \rightarrow PbO$ (C) $Zr \rightarrow ZrO_2$ (D) $Mg \rightarrow MgO$

ML0106

2. Which of the above metal oxide is having minimum thermal decomposition temperature.

(A) CaO (B) FeO (C) ZrO_2 (D) MgO

ML0106

3. Which of the following metal's oxide can be reduced by Fe as reducing agent at temperature (T_1)

(A) Zr (B) Ca (C) Mg (D) None of these

ML0106

PURIFICATION METHOD

Paragraph for 4 to 5

At high temperature carbon reacts with water to produce a mixture of carbon monoxide, CO and hydrogen, H_2.

$$C + H_2O \xrightarrow{\text{red heat}} CO + H_2$$

CO is separated from H_2 and then used to separate nickel from cobalt by forming a volatile compound, nickel tetracarbony, $Ni(CO)_4$.

$$Ni + 4CO \longrightarrow Ni(CO)_4$$

4. How many moles of $Ni(CO)_4$ could be obtained from the CO produced by the reaction of 75.0g of carbon ? Assume 100% reaction and 100% recovery in both steps.

(A) 6.25 (B) 1.563 (C) 3.125 (D) 25.0

ML0107

5. Formation of volatile $Ni(CO)_4$ and its subsequent heating gives pure Ni. Process is called -

(A) Hall (B) Dow (C) Serpeck (D) Mond

ML0107

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MISCELLANEOUS

Match Column

6. Match **Column-I** with **Column-II**

	Column-I (Metals)		**Column-II (Method used for refining)**
(A)	Iron & copper	(P)	Poling
(B)	Zirconium & Titanium	(Q)	Bessemerisation
(C)	Lead & Tin	(R)	Van-Arkel
(D)	Copper & Tin	(S)	Liquation

ML0108

7. Match the following choosing one item from column X and the appropriate item from column Y.

	Column -X		**Column-Y**
(A)	Zinc from $ZnCO_3$	(P)	Calcination
(B)	Lead from PbS	(Q)	Removal of iron
(C)	Cu from $CuFeS_2$	(R)	Froth floatation process
(D)	Tin from cassiterite	(S)	Poling

ML0109

8. Match column (I) (process) with column (II) (electrolyte)

	Column (I) (process)		**Column (II) (electrolyte)**
(A)	Downs cell	(P)	fused $MgCl_2$
(B)	Dow's sea water process	(Q)	fused $(Al_2O_3 + Na_3AlF_6 + CaF_2)$
(C)	Hall-Heroult	(R)	fused (40% NaCl + 60% $CaCl_2$)
		(S)	$(A\ell N + C + N_2)$

ML0110

9. Match column - I with column - II

	Column - I (Property)		**Column - II (Element/compound)**
(A)	Explosive	(P)	Cu
(B)	Self-reduction	(Q)	Fe_3O_4
(C)	Ferrimagnetic material	(R)	$Cu(CH_3COO)_2.Cu(OH)_2$
(D)	Verdigris	(S)	$Pb(NO_3)_2$

ML0111

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10. Match column - I and column - II and select the correct answer using the codes given below the lists:

	Column - I		**Column - II**
(A)	Cyanide process	(P)	Ultrapure Ge
(B)	Floatation process	(Q)	Dressing of HgS
(C)	Electrolytic reduction	(R)	Extraction of Al
(D)	Zone refining	(S)	Extraction of Au

ML0112

11. Match the items of Column I with items of Column II and assign the correct code :

	Column I		**Column II**
(P)	Blistered Cu	(1)	Aluminium
(Q)	Blast furnace	(2)	$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$
(R)	Reverberatory furnace	(3)	Iron
(S)	Hall-Heroult process	(4)	$FeO + SiO_2 \rightarrow FeSiO_3$
		(5)	$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

Code :

(A) P → (2) ; Q → (3) ; R → (4) ; S → (1) (B) P → (1) ; Q → (2) ; R → (3) ; S → (5)

(C) P → (5) ; Q → (4) ; R → (3) ; S → (2) (D) P → (4) ; Q → (5) ; R → (3) ; S → (2)

ML0113

Answer Q.12, Q.13 and Q.14 by appropriately matching the information given in the three columns of the following table.

Column - I Extraction of metal	**Column - II Methods for Reduction**	**Column - III Refining Method**
(1) $SnO_2 \rightarrow Sn$	(i) Carbon Reduction	(P) Poling
(2) $Al_2O_3 \rightarrow Al$	(ii) Hydrometallurgic Reduction	(Q) Electrolytic Refining
(3) $Cu_2S \rightarrow Cu$	(iii) Electrolytic Reduction	(R) Distillation
(4) $ZnS \rightarrow Zn$	(iv) Self-Reduction	(S) Puddling Process

12. Which of the following is **NOT** correctly matched?

(A) (1), (iv), (S) (B) (3), (iv), (P) (C) (4), (i), (R) (D) (2), (iii), (Q)

ML0114

13. Which of the followong match is the **CORRECT** reduction and purification methods for Zn

(A) (i), (R) (B) (iv), (S) (C) (iv), (P) (D) None of these

ML0114

14. Which of the following set of code shows the **CORRECT** similarity with the extraction processes for Pb?

(A) (1), (ii), (S) (B) (4), (iii), (P) (C) (2), (iii), (Q) (D) (3), (iv), (Q)

ML0114

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EXERCISE # JEE-MAINS

1. Which method of purification is represented by the following equation : **[AIEEE-2012]**

$Ti(s) + 2I_2(g) \xrightarrow{523K} TiI_4(g) \xrightarrow{1700K} Ti(s) + 2I_2(g)$

(1) Van Arkel (2) Zone refining (3) Cupellation (4) Poling

ML0119

2. The substance used as froth stabilisers in froth-floatation process is : **[J-Mains-2012 (On line)]**

(1) Copper sulphate (2) Aniline

(3) Sodium cyanide (4) Potassium ethyl xanthate

ML0120

3. Which of the oxide groups among the following cannot be reduced by carbon :- **[J-Mains-2012 (On line)]**

(1) Fe_3O_4, ZnO (2) PbO, Fe_3O_4 (3) Cu_2O, SnO_2 (4) CaO, K_2O

ML0121

4. In Goldschmidt alumino thermic process which of the following reducing agents is used : **[J-Mains-2013 (On line)]**

(1) Calcium (2) Coke (3) Sodium (4) Al-powder

ML0122

5. Calcination is the process in which :

(1) Ore is heated strongly below its melting point in the presence of excess of air and is used for the conversion of carbonates and hydrated oxide ores to their respective oxides.

(2) Ore is heated strongly below its melting point in the absence or limited supply of air and is used for conversion of sulphide ores to their respective oxides

(3) Ore is heated strongly below its melting point either in the limited supply or absence of air and is used to convert carbonates and hydrated oxide ores to their respective oxides

(4) Ore is heated strongly above its melting point in the limited supply of air to convert sulphide ores to their respective oxides.

ML0123

6. The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is : **[JEE-MAINS-2014]**

(1) Cu (2) Cr (3) Ag (4) Ca

ML0124

7. The form of iron obtained from blast furnace is : **[J-Mains-2014 (On line)]**

(1) Steel (2) Wrought Iron (3) Cast Iron (4) Pig iron

ML0125

8. In the context of the Hall-Heroult process for the extraction of Al, which of the following statements is false ? **[JEE-MAINS-2015]**

(1) Al^{3+} is reduced at the cathode to form Al

(2) Na_3AlF_6 serves as the electrolyte

(3) CO and CO_2 are produced in this process

(4) Al_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity

ML0126

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9. Galvanization is applying a coating of :- **[JEE-MAINS-2016]**

(1) Zn (2) Pb (3) Cr (4) Cu

ML0127

10. Which one of the following ores is best concentrated by froth floatation method ? **[JEE-MAINS-2016]**

(1) Malachite (2) Magnetite (3) Siderite (4) Galena

ML0128

11. What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$? **[JEE-MAINS (Online) - 2016]**

(1) The copper metal will dissolve and zinc metal will be deposited

(2) No reaction will occur

(3) The copper metal will dissolve with evolution of oxygen gas

(4) The copper metal will dissolve with evolution of hydrogen gas

ML0129

12. In the leaching method, bauxite ore is digested with a concentrated solution of NaOH that produces 'X'. When CO_2 gas is passed through the aqueous solution of 'X', a hydrated compound 'Y' is precipitated. 'X' and 'Y' respectively are :- **[JEE MAIN ONLINE. 2018]**

(1) $Na[Al(OH)_4]$ and $Al_2O_3 \cdot x\ H_2O$

(2) $Al(OH)_3$ and $Al_2O_3 \cdot x\ H_2O$

(3) $Na[Al(OH)_4]$ and $Al_2(CO_3)_3 \cdot x\ H_2O$

(4) $Na\ AlO_2$ and $Al_2(CO_3)_3 \cdot x\ H_2O$

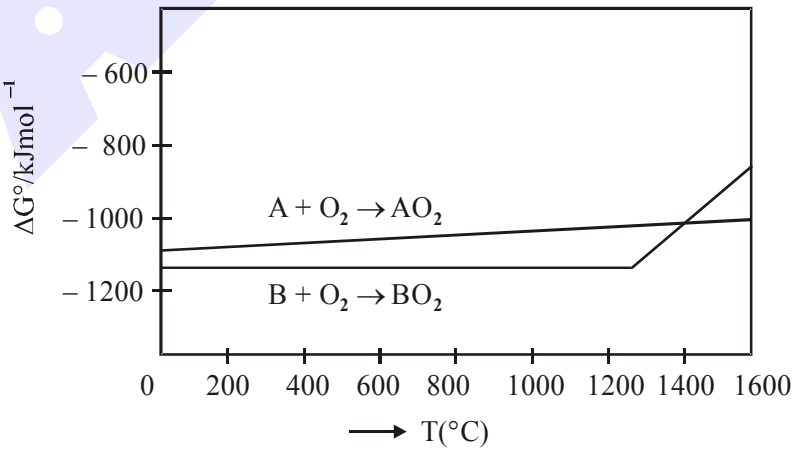
ML0130

13. When metal 'M' is treated with NaOH, a white gelatinous precipitate 'X' is obtained, which is soluble in excess of NaOH. Compound 'X' when heated strongly gives an oxide which is used in chromatography as an adsorbent. The metal 'M' is **[JEE MAIN OFFLINE. 2018]**

(1) Ca (2) Al (3) Fe (4) Zn

ML0131

14. According to the following diagram, A reduces BO_2 when the temperature is : **[J–MAIN 2020]**



(1) < 1400 °C

(2) > 1400 °C

(3) < 1200 °C

(4) > 1200 °C but < 1400 °C

ML0162

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15. The processes of calcination and roasting in metallurgical industries, respectively, can lead to :-
(1) Global warming and acid rain **[J–MAIN 2020]**
(2) Photochemical smog and ozone layer depletion
(3) Global warming and photochemical smog
(4) Photochemical smog and global warming

ML0163

16. Which of the following ore is concentrated using group 1 cyanide salt ? **[J–MAIN 2021]**
(1) Sphalerite (2) Calamine (3) Siderite (4) Malachite

ML0164

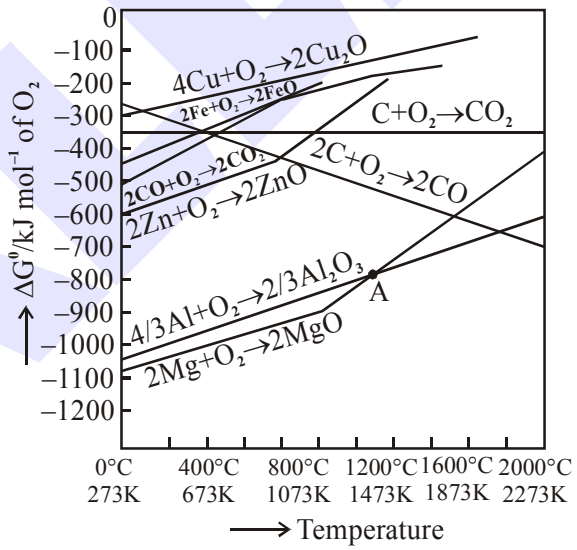
17. Al_2O_3 was leached with alkali to get X. The solution of X on passing of gas Y, forms Z. X, Y and Z respectively are : **[J–MAIN 2021]**
(1) X = $Na[Al(OH)_4]$, Y = SO_2, Z = Al_2O_3 (2) X = $Na[Al(OH)_4]$, Y = CO_2, Z = $Al_2O_3.xH_2O$
(3) X = $Al(OH)_3$, Y = CO_2, Z = Al_2O_3
(4) X = $Al(OH)_3$, Y = SO_2, Z = $Al_2O_3.xH_2O$

ML0165

18. The method used for the purification of Indium is : **[J–MAIN 2021]**
(1) van Arkel method (2) liquation
(3) zone refining (4) vapour phase refining

ML0166

19. The point of intersection and sudden increase in the slope, in the diagram given below, respectively, indicates : **[J–MAIN 2021]**



(1) $\Delta G = 0$ and melting or boiling point of the metal oxide
(2) $\Delta G > 0$ and decomposition of the metal oxide
(3) $\Delta G < 0$ and decomposition of the metal oxide
(4) $\Delta G = 0$ and reduction of the metal oxide

ML0167

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20. In liquation process used for tin (Sn), the metal : **[J-Main 2022]**

(A) is reacted with acid

(B) is dissolved in water

(C) is brought to molten form which is made to flow on a slope

(D) is fused with NaOH.

ML0170

21. Which of the reaction is suitable for concentrating ore by leaching process? **[J-Main 2022]**

(A) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

(B) $Fe_3O_4 + CO \rightarrow 3FeO + CO_2$

(C) $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$

(D) $Al_2O_3 + 6Mg \rightarrow 6MgO + 4Al$

ML0171

22. Among the following ores Bauxite, Siderite, Cuprite, Calamine, Haematite, Kaolinite, Malachite, Magnetite, Sphalerite, Limonite, Cryolite, the number of principal ores if (of) iron is_____. **[J-Main 2022]**

ML0172

23. The metal that is not extracted from its sulphide ore is : **[J-Main 2022]**

(1) Aluminium (2) Iron (3) Lead (4) Zinc

ML0173

24. The compound(s) that is(are) removed as slag during the extraction of copper is :

(A) CaO (B) FeO **[J-Main 2022]**

(C) Al_2O_3 (D) ZnO

(E) NiO

Choose the correct answer from the options given below :

(1) (C) (D) Only (2) (A), (B), (E) Only

(3) (A), (B) Only (4) (B) Only

ML0174

25. Given below are two statements. **[J-Main 2022]**

Statement I : Pig iron is obtained by heating cast iron with scrap iron.

Statement II: Pig iron has a relatively lower carbon content than that of cast iron. In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are not correct.

(C) Statement I is correct but Statement II is not correct

(D) Statement I is not correct but Statement II is correct. **ML0175**

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EXERCISE # JEE-ADVANCED

1. Carnallite does not contain
(A) K (B) Ca (C) Mg (D) Cl

ML0132

2. During initial treatment, preferential wetting of ore by oil and gangue by water takes place in
(A) Levigation (gravity separation) (B) Froth floatation
(C) Leaching (D) Bessemerisation

ML0133

3. Which of the following is true for calcination of a metal ore?
(A) It makes the ore more porous
(B) The ore is heated to a temperature when fusion just begins
(C) Hydrated salts lose their water of crystallisation
(D) Sulphur in sulphides is oxidised to SO_2
(E) Heating with carbon leads to better calcination

ML0134

4. In the commercial electrochemical process for aluminium extraction, the electrolyte used as : **[JEE-1999]**

(A) $Al(OH)_3$ in NaOH solution
(B) an aqueous solution of $Al_2(SO_4)_3$
(C) a molten mixture of Al_2O_3 and Na_3AlF_6
(D) a molten mixture of AlO(OH) and $Al(OH)_3$

ML0135

5. The chemical process in the production of steel from haematite ore involve: **[2000 Qualifying]**
(A) reduction (B) oxidation
(C) reduction followed by oxidation (D) oxidation followed by reduction

ML0136

6. Electrolytic reduction of alumina to aluminium by Hall-Heroult process is carried out:
(A) in the presence of NaCl **[2000 Qualifying]**
(B) in the presence of fluorite
(C) in the presence of cryolite which forms a melt with lower melting temperature
(D) in the presence of cryolite which forms a melt with higher melting temperature

ML0137

7. The chemical composition of "slag" formed during the smelting process in the extraction of copper is : **[2001 Qualifying]**
(A) $Cu_2O + FeS$ (B) $FeSiO_3$ (C) $CuFeS_2$ (D) $Cu_2S + FeO$

ML0138

8. Which of the following processes is used in extractive metallurgy of magnesium? **[2002 Qualifying]**
(A) Fused salt electrolysis (B) Self reduction
(C) Aqueous solution electrolysis (D) Thermite reduction

ML0139

9. In the process of extraction of gold, **[2003 Qualifying]**

$$\text{Roasted gold ore} + CN^- + H_2O \xrightarrow{O_2} [X] + OH^-$$
$$[X] + Zn \longrightarrow [Y] + Au$$

Identify the complexes [X] and [Y] :
(A) $X = [Au(CN)_2]^-$, $Y = [Zn(CN)_4]^{2-}$ (B) $X = [Au(CN)_4]^{3-}$, $Y = [Zn(CN)_4]^{2-}$
(C) $X = [Au(CN)_2]^-$, $Y = [Zn(CN)_6]^{4-}$ (D) $X = [Au(CN)_4]^-$, $Y = [Zn(CN)_4]^{2-}$

ML0140

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10. The methods chiefly used for the extraction of lead and tin from their ores are respectively : **[JEE-2004]**

(A) self reduction and carbon reduction (B) self reduction and electrolytic reduction
(C) carbon reduction and self reduction (D) cyanide process and carbon reduction

ML0141

11. Which ore contains both iron and copper ? **JEE-2004]**

(A) Cuprite (B) Chalcocite (C) Chalcopyrite (D) Malachite

ML0142

12. Extraction for zinc from zinc blende is achieved by : **[JEE-2007]**

(A) electrolytic reduction
(B) roasting followed by reduction with carbon
(C) roasting followed by reduction with another metal
(D) roasting followed by self-reduction

ML0143

13. Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of :-

(A) nitrogen (B) oxygen (C) carbon dioxide (D) argon **[JEE-2008]**

ML0144

Paragraph for questions 14 to 16

Copper is the most nobel of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcanthite ($CuSO_4 . 5H_2O$), atacamite ($Cu_2Cl(OH)_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($Cu_2(OH)_2CO_3$). However, 80% of the world copper production comes from the ore chalcopyrite ($CuFeS_2$). The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction. **[JEE-2010]**

14. Partial roasting of chalcopyrite produces :-

(A) Cu_2S and FeO (B) Cu_2O and FeO (C) CuS and Fe_2O_3 (D) Cu_2O and Fe_2O_3

ML0145

15. Iron is removed from chalcopyrite as :-

(A) FeO (B) FeS (C) Fe_2O_3 (D) $FeSiO_3$

ML0145

16. In self-reduction, the reducing species is :-

(A) S (B) O^{2-} (C) S^{2-} (D) SO_2

ML0145

17. Match the extraction processes listed in column I with metals listed in column II. **[JEE-2006]**

Column I	**Column II**
(A) Self reduction	(P) Lead
(B) Carbon reduction	(Q) Silver
(C) Complex formation and displacement by metal	(R) Copper
(D) Decomposition of iodide	(S) Boron

ML0146

18. Match the conversions in **Column I** with the type(s) of reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. **[JEE-2008]**

Column I	**Column II**
(A) $PbS \rightarrow PbO$	(P) Roasting
(B) $CaCO_3 \rightarrow CaO$	(Q) Calcination
(C) $ZnS \rightarrow Zn$	(R) Carbon reduction
(D) $Cu_2S \rightarrow Cu$	(S) Self reduction

ML0147

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19. In extractive metallurgy of zinc partial fusion of ZnO with coke is called _____ and reduction of the ore to the molten metal is called _____ (smelting, calcining, roasting, sintering). **[JEE-1988]**

ML0148

20. Extraction of metal from the ore cassiterite involves **[JEE-2011]**

(A) carbon reduction of an oxide ore (B) self-reduction of a sulphide ore
(C) removal of copper impurity (D) removal of iron impurity

ML0149

21. Oxidation states of the metal in the minerals haematite and magnetite, respectively, are **[JEE-2011]**

(A) II, III in haematite and III in magnetite (B) II, III in haematite and II in magnetite
(C) II in haematite and II, III in magnetite (D) III in haematite and II, III in magnetite

ML0150

22. In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are : **[JEE-2012]**

(A) O_2 and CO respectively. (B) O_2 and Zn dust respectively.
(C) HNO_3 and Zn dust respectively. (D) HNO_3 and CO respectively.

ML0151

23. Sulfide ores are common for the metals - **[JEE-2013]**

(A) Ag, Cu and Pb (B) Ag, Cu and Sn (C) Ag, Mg and Pb (D) Al, Cu and Pb

ML0152

24. The carbon-based reduction method is **NOT** used for the extraction of **[JEE-2013]**

(A) tin from SnO_2 (B) Iron from Fe_2O_3
(C) aluminium from Al_2O_3 (D) magnesium from $MgCO_3.CaCO_3$

ML0153

25. Upon heating with Cu_2S, the reagent(s) that give copper metal is/are **[JEE Adv. 2014]**

(A) $CuFeS_2$ (B) CuO (C) Cu_2O (D) $CuSO_4$

ML0154

26. Copper is purified by electrolytic refining of blister copper. The correct statement(s) about this process is (are) **[JEE Adv. 2015]**

(A) Impure Cu strip is used as cathode
(B) Acidified aqueuous $CuSO_4$ is used as electrolyte
(C) Pure Cu deposits at cathode
(D) Impurities settle as anode-mud

ML0155

27. Match the anionic species given in Column-I that are present in the ore(s) given in Column-II **[JEE Adv. 2015]**

Column - I	**Column - II**
(A) Carbonate	(P) Siderite
(B) Sulphide	(Q) Malachite
(C) Hydroxide	(R) Bauxite
(D) Oxide	(S) Calamine
	(T) Argentite

ML0156

28. Extraction of copper from copper pyrite ($CuFeS_2$) involves **[JEE Adv. 2016]**

(A) crushing followed by concentration of the ore by froth-flotation
(B) removal of iron as slag
(C) self-reduction step to produce 'blister copper' following evolution of SO_2
(D) refining of 'blister copper' by carbon reduction

ML0157

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29. Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnance such that the contents undergo self-reduction. The weight (in kg) of Pb produced per kg of O_2 consumed is _____ . (Atomic weights in g mol^{-1} : O = 16, S = 32, Pb = 207) **[JEE ADV. 2018]**

ML0158

30. Calamine, malachite, magnetite and cryolite, respectively are **[JEE ADV. 2019]**

(1) $ZnSO_4$, $CuCO_3$, Fe_2O_3 , AlF_3

(2) $ZnCO_3$, $CuCO_3.Cu(OH)_2$, Fe_3O_4, Na_3AlF_6

(3) $ZnSO_4$, $Cu(OH)_2$, Fe_3O_4, Na_3AlF_6

(4) $ZnCO_3$, $CuCO_3$, Fe_2O_3, Na_3AlF_6

ML0159

31. The cyanide process of gold extraction involves leaching out gold from its ore with CN^- in the presence of **Q** in water to form **R**. Subsequently, **R** is treated with **T** to obtain Au and **Z**. Choose the correct option(s). **[JEE ADV. 2019]**

(1) **T** is Zn (2) **R** is $[Au(CN)_4]^-$ (3) **Z** is $[Zn(CN)_4]^{2-}$ (4) **Q** is O_2

ML0160

32. Which among the following statement(s) is(are) true for the extraction of aluminium from bauxite? **[JEE ADV. 2020]**

(A) Hydrated Al_2O_3 precipitates, when CO_2 is bubbled through a solution of sodium aluminate.

(B) Addition of Na_3AlF_6 lowers the melting point of alumina.

(C) CO_2 is evolved at the anode during electrolysis.

(D) The cathode is a steel vessel with a lining of carbon.

ML0161

33. The correct statement(s) related to the metal extraction processes is(are) **[JEE ADV. 2021]**

(A) A mixture of PbS and PbO undergoes self-reduction to produce Pb and SO_2.

(B) In the extraction process of copper from copper pyrites, silica is added to produce copper silicate.

(C) Partial oxidation of sulphide ore of copper by roasting, followed by self-reduction produces blister copper.

(D) In cyanide process, zinc powder is utilized to precipitate gold from $Na[Au(CN)_2]$

ML0169

34. The correct option(s) related to the extraction of iron from its ore in the blast furnace operating in the temperature range 900 - 1500 K is(are) **[JEE ADV. 2022]**

(A) Limestone is used to remove silicate impurity.

(B) Pig iron obtained from blast furnace contains about 4% carbon.

(C) Coke (C) converts CO_2 to CO.

(D) Exhaust gases consist of NO_2 and CO.

ML0176

35. The electrochemical extraction of aluminum from bauxite ore involves. **[JEE ADV. 2022]**

(A) The reaction of Al_2O_3 with coke (C) at a temperature > 2500°C.

(B) The neutralization of aluminate solution by passing CO_2 gas to precipitate hydrated alumina ($Al_2O_3.3H_2O$)

(C) The dissolution of Al_2O_3 in hot aqueous NaOH.

(D) the electrolysis of Al_2O_3 mixed with Na_3AlF_6 to give Al and CO_2.

ML0177

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ANSWER-KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	B	B	C	A	A	B	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	B	B	A	D	A	A	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	C	B	B	D	C	B	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	B	C	B	A	C	C	D	A	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	A	A	D	C	C	C	B	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	C	B	D	B	A	D	D	C	D
Que.	61	62	63	64	65	66				
Ans.	B	B	A	A	B	C				

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,D	A,B,C	B,C	A,B	A,C,D	A,B,C,D	A,C	A,C	A,B	A,B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B	B,C	A,B	B,C	A,B,C,D	A,B	A,B,C	B,C,D	A,B	A,B,C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C,D	A,D	A,B	A,D	A,B,D	B,D	B,C	C,D	A,C	B,C

EXERCISE # S-1

Que.	1	2	3	4	5	6	7	8	9	
Ans.	3	3	4	4	2	4	4	4	2	

EXERCISE # S-2

1	2	3	4	5	6
B	C	A	B	D	A→Q ; B→R ; C→S ; D→P
7					8
A→P ; B→R ; C→Q,R,S ; D→Q,S					A→R ; B→P ; C→Q
9					10
A→S ; B→P ; C→Q ; D→R					A→S ; B→Q ; C→R; D→P
11	12	13	14		
A	A	A	D		

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EXERCISE # JEE-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	4	4	3	4	4	2	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	2	1	1	2	3	1	C
Que.	21	22	23	24	25					
Ans.	C	4	1	4	B					

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	A,C	C	C	C	B	A	A	A
Que.	11	12	13	14	15	16	17			
Ans.	C	B	B	A	D	C	A→P,R ; B→P ; C→Q ; D→S			
Que.	18									
Ans.	A→P ; B→Q ; C→P,R ; D→P,S									

19. Sintering , Smelting

Que.	20	21	22	23	24	25	26	27	
Ans.	A,D or A,C,D	D	B	A	C,D	B,C,D	B,C,D	A→P,Q,S ; B→T ; C→Q,R ; D→R	
Que.	28	29	30	31	32	33	34	35	
Ans.	A,B,C	6.47	2	1,3,4	A,B,C,D	A,C,D	A,B,C	B,C,D	

29. $PbS + O_2 \longrightarrow Pb + SO_2$

$\frac{1000}{32}$ mol $\qquad \frac{1000}{32} \times 207$ gm

mol of Pb = mol of O_2

$= \frac{1000}{32}$ mol

$\therefore$ mass of Pb $= \frac{1000}{32} \times 207$g

$= \frac{207}{32}$ kg $= 6.47$ kg

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